

Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

Dr.-Ing. Dominik Brands, Julia Sunten, M.Sc.

Prof. Dr.-Ing. habil. Jörg Schröder
Institut für Mechanik
Abteilung Bauwissenschaften
Fakultät für Ingenieurwissenschaften
Universität Duisburg-Essen

Exercise 5: Contact Simulation without Contact Wizard

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

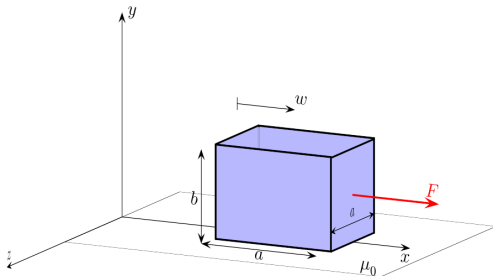
Contact Modeling in ANSYS without using the Contact Wizard

- The contact wizard always chooses contact elements of type CONTA171 / CONTA174 and TARGE169 / TARGE170, respectively.
- ANSYS provides additional contact elements that can only be used *directly*:
e.g. CONTA175
(remark: CONTA175 replaces older contacts CONTAC26, CONTAC48, CONTAC52)
- Ways to improve the results of a contact simulation:
 - Using different types of contact elements
 - Varying the contact stiffness
 - Modifying the mesh
 - Changing the solution method (Lagrange, Penalty, Augmented-Lagrange)
 - Generating symmetric contact

View contact status before solving: CNCHECK,POST

Simulation of a shifted box

Problem description



- Geometric data

$$a = 300 \text{ mm}$$

$$b = 150 \text{ mm}$$

$$\text{depth} = 200 \text{ mm}$$

- Material data

Box (Aluminium):

$$E = 72,200 \text{ N/mm}^2$$

$$\nu = 0.33$$

$$\rho = 2.7 \text{ g/cm}^3$$

Base is supposed to be rigid

- Coefficient of static friction

$$\mu_0 = 0.2$$

- Load

$$\text{Gravity: } g = 9.81 \text{ m/s}^2$$

$$\text{Displacement: } w = 600 \text{ mm}$$

- Objective target

$$\text{Friction force } F$$

Aspects of Modeling

Simulation of a shifted box

- Writing input files (APDL):
 - `model.dat`: pre processing (material, element type, meshing)
 - `load.dat`: solution (load, boundary condition, solver)
 - `post.dat`: post processing (reporting of displacement, reaction force)
- Generating a parametric FE-model w.r.t. number of elements
- Contact modeling *without* employing the *Contact Wizard*
- Including gravitational effects
- Advanced postprocessing (calculation of reaction force)
- Createing animations
- Modifying the model:
 - Testing different (contact) elements
 - Altering the mesh
 - Varying the contact stiffness and the penetration



3D model, boundary condition, load

Simulation of a shifted box

• Element types

- 2-D PLANE182
(Simplified enhanced strain formulation)
(plane-stress-condition with thickness)
- CONTA175, 2-D/3-D Node-to-Surface Contact
- TARGE169, 2-D Target Element

• Boundary conditions

Nodes	Coordinates	Boundary conditions
Na	(0, 0)	$u_x = 0, u_y = 0$
Nb	(1000, 0)	$u_x = 0, u_y = 0$
N(L1)	$(200 + \Delta l, 0 < y < 150)$	$u_x = w$

• Meshing/Modeling

Creating an input file `model.dat`:

Parametric model (number of elements is adjustable)

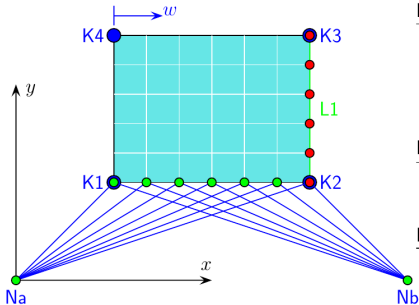
• Load steps and post processing

Creating an input file `load.dat`:

1. LS: applying the gravitational force
2. LS: shifting the box (L1) by $w = 600$ mm

Creating an input file `post.dat`:

Summation of reaction forces (at nodes of L1)



Additional ANSYS APDL Commands

Simulation of a shifted box

Pre Processing

/PREP7

Definition of friction coefficient

TB, ...
TBDATA, ...

Controlling additional element parameter:

KEYOPT, ... (via Key-Option)

- 2D approximation type
- contact stiffness automatically

Controlling additional element (real) constants:

R, ... (via real constant)

- thickness for plane-stress-condition (for PLANE182)
- contact stiffness manually (for CONTA175 & TARGET169)

Activating the contact element type

TYPE, ...

Activating the corresponding real constant set

REAL, ...

Select the target geometry

TSHAP, ...

Generating of target elements

LMESH, ...

Generating of contact elements (as overlay on existing nodes)

ESURF, ...

more help: help,XXX, whereby XXX should be replace by command.

Additional ANSYS APDL Commands (contd.)

Simulation of a shifted box

Post Processing

`/POST26`

Specifies the total reaction force data to be stored

`RFORCE,...`

Add variables

`ADD,...`

Multiplies variables

`PROD,...`

Retrieves a value and stores it

`*GET,...`

e.g. `*GET,num,NODE,0,NUM,MAX` → highest node number in the selected set

e.g. `*GET,y_max,NODE,0,MXLOC,Y` → max. nodal y-coord in the selected set

e.g. `*GET,numnodes,NODE,0,COUNT` → number of nodes in the selected set

more help: `help,XXX`, whereby XXX should be replace by command.